CHROMATIC HOMOTOPY THEORY SEMINAR MÜNSTER

Disclaimer: You can follow these notes during the talk or extend them or prepare your own. Essential references and content to be covered are always written at the beginning of each talk. All gray text is optional and can be easily skipped as it only provides some additional or background material which simplifies the understanding (in the author's opinion).

1. Talk 4. Complex Bordism and its Universal Property

Main reference [1]; for Thom spectra see also [3, §10].

- Motivation: a complex orientation [Definition 9.1] is a homotopy class of maps $\Sigma^{\infty-2}\mathbb{CP}^\infty \to E$ sending the bottom cell to the unit; we look for a homotopy ring spectrum corepresenting orientations [Remark 9.3]. Homotopy ring spectra versus E_∞-rings.
- Heuristic: the pointed symmetric algebra $\widetilde{\mathrm{Sym}}(V)$ and its universal property (the paragraph after [Remark 9.3]); a complex-oriented theory "thinks" that $BU(n)$ is $\mathrm{Sym}^n \mathbb{CP}^\infty$ (the paragraph before [Remark 9.4]).
- Thom spaces classically [Definition 9.5], [Example 9.6], and as colimits of sphere-valued functors [3, Def. 10.1, 10.3, Ex. 10.2, 10.5]. Un/straightening; comparison of the two definitions [Remark 9.7]. Thom spectra of virtual bundles.
- Define $MU(n)$ [Notation 9.9] and prove [Lemma 9.8]. Then [Remark 9.11]: $MU(0) \simeq \mathbb{S}$, $MU(1) \simeq \Sigma^{\infty-2}\mathbb{CP}^\infty$, so an orientation is a map $MU(1) \to E$ restricting to the unit. The maps $j_n \colon MU(n-1) \to MU(n)$, [Remark 9.10] and Thom classes.
- Define MU [Definition 9.12] and its homotopy ring structure (the paragraph after [Definition 9.12], [2, Lemma 4.4.4]). State [Theorem 9.14] and prove that every orientation comes from a ring map $MU \to E$, using the Milnor sequence. Examples: $H\mathbb{Z}$ and the Todd genus. The E_∞-structure [Remark 9.13] and its universal property [4, Rem. 2.33], [6].
- All ring maps $MU \to E$ via changes of orientation. State [Lemma 11.1] and its reformulation [Remark 11.2].
- State Quillen's theorem [Theorem 9.15 = Theorem 11.8].
- Outlook: MU_*MU and [Theorem 11.10]; the Čech nerve of $\mathbb{S} \to MU$ as the topological shadow of $\operatorname{Spec} L \to \mathcal{M}_{\mathrm{fg}}$ (continued in Talk 5).

In Talk 1 we saw that a multiplicative cohomology theory E with a complex orientation produces a formal group law F_E over the coefficient ring E^*. In Talk 3 we showed that formal group laws are classified by maps from the Lazard ring L. Thus, we have a ring map $L \to E^*$. Today we construct the *universal complex-oriented theory* (MU, t_{MU}), which will be initial among complex-oriented theories.

Conventions/Reminder: Sp is the stable ∞-category of spectra with its symmetric monoidal structure $\otimes$, commuting with colimits in each variable. The unit is given by the sphere spectrum $\mathbb{S}$. Denote by $h\mathrm{Sp}$ the homotopy category of Sp.

For spectra E and X we write

$$E^n(X) = [X, \Sigma^n E] = \pi_{-n} \operatorname{map}(X, E), \qquad E_n(X) = \pi_n(E \otimes X).$$

For a (pointed) space Y we set $E^n(Y) = E^n(\Sigma^\infty Y)$, which agrees with the classical definition.

Definition 1.1. A *homotopy ring spectrum* is a commutative monoid in $(h\mathrm{Sp}, \otimes, \mathbb{S})$, i.e. a spectrum E with a unit $\eta \colon \mathbb{S} \to E$ and a multiplication $\mu \colon E \otimes E \to E$ which are unital, associative and

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commutative *up to homotopy*. A map of homotopy ring spectra is a map in $h\mathrm{Sp}$ commuting with η and μ up to homotopy.

Remark 1.2. In the first talk we worked with multiplicative cohomology theories; those are precisely the cohomology theories represented by homotopy ring spectra.

Definition 1.3. An *E_∞-ring* is a commutative algebra in the ∞-category Sp; the homotopies witnessing associativity and commutativity are part of the data, together with all higher coherences between them.

Recall the definition:

Definition 1.4. Let E be a homotopy ring spectrum. A *complex orientation* of E is a class $t \in \widetilde{E}^2(\mathbb{CP}^\infty)$ whose restriction to $\widetilde{E}^2(\mathbb{CP}^1) \cong \widetilde{E}^2(S^2) \cong E^0(*)$ is the unit 1.

Now we're reformulating the definition of a complex orientation in terms of spectra. Note that $\widetilde{E}^2(\mathbb{CP}^\infty) = [\Sigma^\infty \mathbb{CP}^\infty, \Sigma^2 E] \cong [\Sigma^{\infty-2}\mathbb{CP}^\infty, E]$. Thus, a choice of orientation t is a choice of a homotopy class of maps $t\colon \Sigma^{\infty-2}\mathbb{CP}^\infty \to E$ such that

$$i\colon \mathbb{S} \cong \Sigma^{\infty-2}S^2 \cong \Sigma^{\infty-2}\mathbb{CP}^1 \longrightarrow \Sigma^{\infty-2}\mathbb{CP}^\infty \xrightarrow{t} E$$

gives (a homotopy class of) the unit of a homotopy ring spectrum E.

Brand-new definition 1.5. A complex orientation of E is a homotopy class of maps of spectra $t\colon \Sigma^{\infty-2}\mathbb{CP}^\infty \to E$ such that $t \circ i = \eta$ is the unit of E.

So complex orientations are maps out of a fixed spectrum $\Sigma^{\infty-2}\mathbb{CP}^\infty$ which is *pointed* by i, and which have to send the point to the unit. Our goal:

find a homotopy ring spectrum MU such that ring maps $MU \to E$ are the same as complex orientations of E.

To gain some intuition for the construction of MU, it might be helpful to think about the following classical story:

Let V be a vector space over a field k, pointed by $u\colon k \to V$. Multiplication by u gives maps $\mathrm{Sym}^{n-1}(V) \to \mathrm{Sym}^n(V)$ (hence all maps of the form $V^{\otimes(n-1)} \to V^{\otimes n}$ induced by applying u in one of the coordinates are identified), and we put

$$\widetilde{\mathrm{Sym}}(V) := \mathrm{colim}\big(k = \mathrm{Sym}^0 V \xrightarrow{u} \mathrm{Sym}^1 V \xrightarrow{u} \mathrm{Sym}^2 V \xrightarrow{u} \cdots\big).$$

Lemma 1.6. *For any commutative k-algebra A,*

$$\mathrm{Hom}_{\mathrm{CAlg}_k}\big(\widetilde{\mathrm{Sym}}(V), A\big) \cong \{f\colon V \to A \text{ linear} \mid f(u) = 1\}.$$

So we want "$MU = \widetilde{\mathrm{Sym}}(\Sigma^{\infty-2}\mathbb{CP}^\infty)$" in some sense.

Recall that for a complex-oriented E,

$$E^*(BU(n)) \cong \big(E^*(\mathbb{CP}^\infty)^{\otimes n}\big)^{\Sigma_n},$$

so every complex-oriented theory "thinks" that $BU(n)$ is the n-th symmetric power of $\mathbb{CP}^\infty = BU(1)$.

We will therefore build MU out of the spaces $BU(n)$, suitably shifted so that the $n = 1$ term is $\Sigma^{\infty-2}\mathbb{CP}^\infty$ (which is V in our analogy) and the $n = 0$ term is $\mathbb{S}$ (which is the basepoint). The tool for this shifting is the Thom spectrum.

Classically. Let V be a real vector bundle of rank n over a CW-complex X, with a metric.

Definition 1.7. The *Thom space* of V is $\mathrm{Th}_X(V) := D(V)/S(V)$, the unit disc bundle with the unit sphere bundle collapsed to a point. It is canonically pointed.

Example 1.8. For the trivial bundle $V = X \times \mathbb{R}^n$ we get $(X \times D^n)/(X \times S^{n-1}) \cong \Sigma^n X_+$. Since every vector bundle is locally trivial, a Thom space should be thought of as a *twisted suspension* of X_+.

Theorem 1.9 (Un/straightening)**.** *Let B be an anima. There is an equivalence of ∞-categories*

$$\mathrm{Fun}(B, \mathrm{Cat}_\infty) \simeq \mathrm{coCart}_B \subset \mathrm{Cat}_{\infty/B}$$

between functors and coCartesian fibrations over B. Furthermore, for an object $(E \to B) \in \mathrm{coCart}_B$ with straightening $F_E \colon B \to \mathrm{Cat}_\infty$, $b \mapsto E_b$, we have an equivalence $E \cong \mathrm{colim}_B F_E$, i.e. E assembles from the fibers over B.

It gives a clue for the next modern definition of Thom spaces/spectra.

Definition 1.10. Let B be an anima.

(1) For a functor $F \colon B \to \mathrm{An}_*$ taking values in pointed spaces equivalent to spheres, the *Thom space* is $\mathrm{Th}(B, F) := \mathrm{colim}_B F$, the colimit taken in pointed anima An_*.
(2) For a functor $F \colon B \to \mathrm{Sp}$ taking values in spectra equivalent to $\mathbb{S}^n$, $n \in \mathbb{Z}$, the *Thom spectrum* is $\mathrm{th}(B, F) := \mathrm{colim}_B F$, the colimit taken in Sp.

Example 1.11.

(1) If F is constant with value $\mathbb{S}$, then $\mathrm{th}(B, F) = \mathrm{colim}_B \mathbb{S} \cong \mathbb{S} \otimes \Sigma^\infty_+ B \cong \Sigma^\infty_+ B$. In spaces, a constant functor with value S^n gives $\mathrm{Th} = S^n \wedge B_+ = \Sigma^n B_+$ [3, Ex. 10.2].
(2) Since $\Sigma^\infty \colon \mathrm{An}_* \to \mathrm{Sp}$ preserves colimits, $\mathrm{th}(B, \Sigma^\infty F) \simeq \Sigma^\infty \mathrm{Th}(B, F)$ [3, Ex. 10.5].

To construct functors of this form, consider any real vector bundle V over X.

It is classified by a functor $X \to \mathrm{Vect}^{\simeq}_{\mathbb{R}} \simeq \coprod_n BO(n)$ into the ∞-groupoid of finite-dimensional real vector spaces and their isomorphisms, and one-point compactification is a functor $\mathrm{Vect}^{\simeq}_{\mathbb{R}} \to \mathrm{An}_*$ with values in spheres. Composing, we get $S^V \colon X \to \mathrm{An}_*$.

In fact, our fancy definition agrees with a historical one in this case.

Proposition 1.12 ([Remark 9.7] [1])**.** $\mathrm{Th}(X, S^V) \simeq \mathrm{Th}_X(V)$.

Remark 1.13 (Virtual bundles)**.** Since $S^{V \oplus \mathbb{R}^k} \simeq S^V \wedge S^k \cong \Sigma^k S^V$ fiberwise and colimits commute with Σ^k, we have $\mathrm{Th}_X(V \oplus \mathbb{R}^k) \simeq \Sigma^k \mathrm{Th}_X(V)$. So adding trivial bundles is invertible after passing to spectra, and we may define, for a *virtual* bundle $V - \mathbb{R}^k$ (just a formal difference),

$$\mathrm{th}(V - \mathbb{R}^k) := \Sigma^{-k}\Sigma^\infty \mathrm{Th}_X(V) = \mathrm{colim}_X \Sigma^{-k}\Sigma^\infty S^V.$$

This is a colimit of a functor $X \to \mathrm{Sp}$ with values in spheres.

Now we are ready to define the building blocks of our dream spectrum MU, namely the spectra $MU(n)$. As in the first talk, let γ_n be the tautological complex rank n bundle over $BU(n)$, which is a real bundle of rank $2n$.

Definition 1.14 ([Notation 9.9] [1])**.** $MU(n) := \mathrm{th}(\gamma_n - \mathbb{R}^{2n}) = \Sigma^{-2n}\Sigma^\infty \mathrm{Th}_{BU(n)}(\gamma_n)$.

These Thom spaces have a simple description as follows:

Lemma 1.15 ([Lemma 9.8] [1])**.** $\mathrm{Th}_{BU(n)}(\gamma_n) \simeq BU(n)/BU(n-1)$*, the cofiber of the map $BU(n-1) \to BU(n)$ classifying $\gamma_{n-1} \oplus \mathbb{C}$.*

Proof. Recall that for the tautological bundle γ_n we have a concrete model, namely, $\gamma_n = \{(V, v) \mid V \text{ an } n\text{-dimensional complex vector space}, v \in V\}$. Each fiber is a complex vector space, so it has a metric, and we see that the disc bundle $D(\gamma_n)$ can be retracted to the zero $0 \in V$. In other words, the disc bundle $D(\gamma_n)$ deformation retracts onto its zero section, so $D(\gamma_n) \simeq BU(n)$.

The sphere bundle $S(\gamma_n)$ in this model is given by $\{(V, v) \mid |v| = 1\}$. We will prove that $S(\gamma_n) \simeq BU(n-1)$. Consider the map $S(\gamma_n) \to BU(n-1)$ sending $(V, v) \mapsto v^\perp$, the orthogonal complement of v in V. The fiber of this map over $W \in BU(n-1)$ is the unit sphere in the complement of W in $\mathbb{C}^\infty$, which is homeomorphic to S^∞, since the complement is homeomorphic to $\mathbb{C}^\infty$. Thus, the fiber is contractible, and the map is a homotopy equivalence as both spaces can be realized as CW-complexes.

Since $S(\gamma_n) \to D(\gamma_n)$ is a cellular inclusion, the cofiber of this map is homotopy equivalent to the homotopy cofiber $BU(n)/BU(n-1)$. □

Corollary 1.16 ([Remark 9.11] [1])**.** *$MU(0) \simeq \mathbb{S}$ and $MU(1) \simeq \Sigma^{\infty-2}\mathbb{CP}^\infty$.*

Proof. $BU(0) = \text{pt}$ and $BU(1) = \mathbb{CP}^\infty$, so $\text{Th}(\gamma_1) \simeq \mathbb{CP}^\infty$. □

As a next step, we want to construct morphisms $MU(n-1) \to MU(n)$. There is a pullback diagram, giving a morphism of bundles:

$$\begin{array}{ccc} \gamma_{n-1} \oplus \mathbb{C} & \longrightarrow & \gamma_n \\ \downarrow & & \downarrow \\ BU(n-1) & \longrightarrow & BU(n) \end{array}$$

It can be easily seen, let's say, from explicit models: the map $BU(n-1) \to BU(n)$ sends an $(n-1)$-dimensional space V in $\mathbb{C}^\infty$ to the n-dimensional space $s(V) \oplus \mathbb{C}e_1$, where s is the operator which shifts basis vectors $e_i \mapsto e_{i+1}$. Hence an explicit description of the pullback bundle over $BU(n-1)$ is $\{(W, v) \mid v \in s(W) \oplus \mathbb{C}e_1\} \cong \gamma_{n-1} \oplus \mathbb{C}$.

Thus, we obtain a map between the Thom spaces $\text{Th}_{BU(n-1)}(\gamma_{n-1} \oplus \mathbb{C}) \simeq \Sigma^2 \,\text{Th}_{BU(n-1)}(\gamma_{n-1}) \to \text{Th}_{BU(n)}(\gamma_n)$. Applying Σ^∞ and shifting by $-2n$, we obtain

$$j_n \colon MU(n-1) = \Sigma^{-2(n-1)}\Sigma^\infty \,\text{Th}_{BU(n-1)}(\gamma_{n-1}) \longrightarrow \Sigma^{-2n}\Sigma^\infty \,\text{Th}_{BU(n)}(\gamma_n) = MU(n).$$

Remark 1.17. Under these equivalences the map $MU(0) \to MU(1)$ is $i \colon \mathbb{S} \to \Sigma^{\infty-2}\mathbb{CP}^\infty$. Hence a complex orientation of E is the same as a map $MU(1) \to E$ whose restriction to $MU(0) = \mathbb{S}$ is the unit.

Now let E be complex oriented. Restriction $E^*(BU(n)) \to E^*(BU(n-1))$ sends $c_i \mapsto c_i$ for $i < n$ and $c_n \mapsto c_n(\gamma_{n-1} \oplus \mathbb{C}) = c(\gamma_{n-1})_n = 0$, so it is surjective with kernel (c_n).

The long exact sequence of the cofiber sequence of Lemma 1.15 therefore splits into short exact sequences, and we get:

Proposition 1.18 ([Remark 9.10] [1])**.** *For complex-oriented E,*

$$E^*(\text{Th}_{BU(n)} \gamma_n) \cong c_n \cdot E^*[[c_1, \ldots, c_n]] \subseteq E^*(BU(n)).$$

Note that

$$(1)\quad E^*(MU(n)) = \pi_* \,\text{map}(MU(n), E) \cong \pi_*(\text{Map}(\text{Th}_{BU(n)}(\gamma_n), \Omega^{2n}\Omega^\infty E)) \cong E^{*+2n}(\text{Th}_{BU(n)}(\gamma_n)).$$

Therefore, the previous computation endows $E^*(MU(n))$ with a structure of a free $E^*(BU(n))$-module of rank one on the *Thom class* $u_n \in E^0(MU(n))$ corresponding to c_n under the isomorphism above.

Under these isomorphisms, $j_n^*\colon E^*(MU(n)) \to E^*(MU(n-1))$ corresponds to the restriction $E^*(BU(n)) \to E^*(BU(n-1))$; in particular j_n^* is surjective.

Similarly,

$$E^*(MU(n) \otimes MU(m)) \cong E^*(BU(n)) \otimes E^*(BU(m)) \cdot (u_n \otimes u_m) \subset E^*(BU(n) \times BU(m)), \tag{2}$$

and the restriction maps in n and m are surjective. For u_1 we get $c_1 = t$: the Thom class of $MU(1) = \Sigma^{\infty-2}\mathbb{CP}^\infty$ is the orientation itself.

Finally, we are defining the main object of interest, the complex bordism spectrum MU.

Definition 1.19 ([Definition 9.12] [1])**.** The *complex bordism spectrum* is the colimit in Sp

$$MU := \operatorname{colim}\big(\mathbb{S} = MU(0) \xrightarrow{j_1} MU(1) \xrightarrow{j_2} MU(2) \xrightarrow{j_3} \cdots\big).$$

We write $t\colon \Sigma^{\infty-2}\mathbb{CP}^\infty = MU(1) \to MU$ for the canonical map.

To define the homotopy ring structure, we take the unit to be $\eta\colon \mathbb{S} = MU(0) \to MU$. For the multiplication, the direct sum map $\oplus\colon BU(n) \times BU(m) \to BU(n+m)$ pulls γ_{n+m} back to the sum $\gamma_n \oplus \gamma_m$ of tautological bundles. Thom spaces take sums to the smash product [2, Lemma 4.4.4] (for the shifts see Remark 1.13), so we get maps

$$\operatorname{th}(V \oplus W) \simeq \Sigma^\infty \operatorname{Th}(V \oplus W) \simeq \Sigma^\infty(\operatorname{Th}(V) \wedge \operatorname{Th}(W)) \simeq \Sigma^\infty \operatorname{Th}(V) \otimes \Sigma^\infty \operatorname{Th}(W) \simeq \operatorname{th}(V) \otimes \operatorname{th}(W),$$

So we obtain maps

$$\mu_{n,m}\colon MU(n) \otimes MU(m) \longrightarrow MU(n+m),$$

compatible with the j's.

Passing to the colimit in Sp (since $MU \otimes MU \simeq \operatorname{colim}_{n,m} MU(n) \otimes MU(m)$ using that the tensor product preserves colimits in each variable), we get $\mu\colon MU \otimes MU \to MU$. Associativity, commutativity and unitality up to homotopy are inherited from the corresponding properties of $\oplus$.

Proposition 1.20. *(MU, η, μ) is a homotopy ring spectrum, and $t\colon MU(1) \to MU$ is a complex orientation of MU.*

Proof. The first part was explained above. For the second, $t \circ i = t \circ j_1 = \eta$ by construction (Remark 1.17). □

Finally, we are ready to show the universal property of the newly constructed spectrum MU.

Theorem 1.21 ([Theorem 9.14] [1])**.** *For any homotopy ring spectrum E, the assignment $f \mapsto f \circ t$ is a bijection*

$$\operatorname{Hom}_{\mathrm{CAlg}(h\mathrm{Sp})}(MU, E) \xrightarrow{\sim} \{\textit{complex orientations of } E\}.$$

Proof. The map is defined as follows: if f is a ring map, then $f \circ t \circ i = f \circ \eta_{MU} = \eta_E$, so $f \circ t$ is an orientation. Let us now prove that it is in fact bijective.

Fix an orientation t_E of E. By Talk 1 it determines Chern classes $c_i \in E^{2i}(BU(n))$, natural and satisfying the Whitney sum formula, with $c_1(\gamma_1) = t_E$ and $E^*(BU(n)) = E^*[[c_1, \dots, c_n]]$. Let $u_n \in E^0(MU(n))$ be the Thom class corresponding to c_n under Proposition 1.18. Note that u_n belongs to the 0-th cohomology of $MU(n)$, reflecting the shift $-2n$ in (1).

Each complex orientation only knows about the map from $MU(1)$, compatible with the unit, so to show bijectivity we need to reconstruct maps from all $MU(n)$ and then from MU. Let us start by noting that it suffices to only construct compatible maps from $MU(n)$:

By the discussion after Proposition 1.18 the maps j_n^* are surjective, so the $\varprojlim^1$-terms in the Milnor sequence vanish and

$$E^*(MU) \cong \varprojlim_n E^*(MU(n)), \qquad E^*(MU \otimes MU) \cong \varprojlim_n E^*(MU(n) \otimes MU(n)),$$

the second using the cofinal diagonal subsystem of $(MU(n) \otimes MU(m))_{n,m}$.

That is, maps out of MU and $MU \otimes MU$ are determined by their restrictions to the $MU(n)$ and $MU(n) \otimes MU(n)$, and every compatible family extends.

Since the fixed orientation gives Chern classes, it also gives Thom classes, and we can let $f_n \colon MU(n) \to E$ be the Thom class u_n. Then $f_0 = 1 = \eta_E$ and $f_1 = t_E$, since $c_1 = t_E$.

We claim that the square

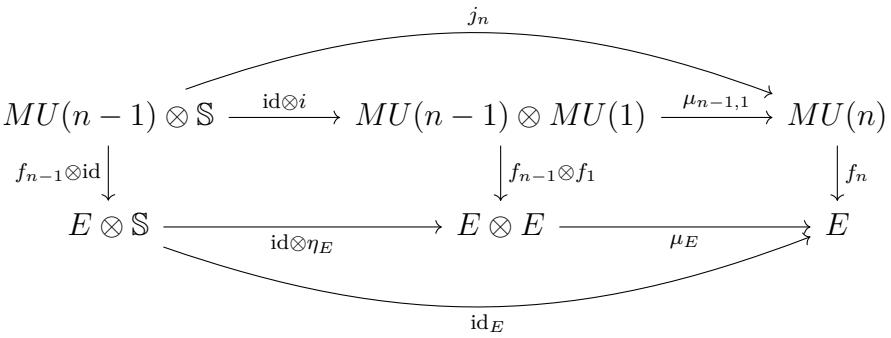
$$\begin{array}{ccc} MU(n) \otimes MU(m) & \xrightarrow{\mu_{n,m}} & MU(n+m) \\ {\scriptstyle f_n \otimes f_m}\downarrow & & \downarrow{\scriptstyle f_{n+m}} \\ E \otimes E & \xrightarrow{\mu_E} & E \end{array}$$

commutes up to homotopy.

Again by the discussion after Proposition 1.18 both composites are classes in $E^0(MU(n) \otimes MU(m)) \cong E^*(BU(n) \times BU(m)) \cdot (u_n \otimes u_m)$. The lower composite is $u_n \otimes u_m$, corresponding to $c_n(\gamma_n)\, c_m(\gamma_m)$; the upper one is the pullback of u_{n+m} along $\oplus$, corresponding to $c_{n+m}(\gamma_n \oplus \gamma_m)$. These agree by the Whitney sum formula.

The maps given by the Thom classes are also compatible with $j_n \colon MU(n-1) \to MU(n)$.

The map j_n factors as $MU(n-1) \simeq MU(n-1) \otimes MU(0) \xrightarrow{\mathrm{id} \otimes i} MU(n-1) \otimes MU(1) \xrightarrow{\mu_{n-1,1}} MU(n)$, because $\gamma_{n-1} \oplus \mathbb{C}$ is the pullback of $\gamma_{n-1} \oplus \gamma_1$ along $BU(n-1) \times \mathrm{pt} \to BU(n-1) \times BU(1)$. Thus, this compatibility follows from the previous square:



The left square commutes since $f_1 \circ i = \eta_E$, the right square is the previous one, and the bottom composite is the identity by unitality.

Using compatibility with multiplication and $f_1 \circ i = \eta_E$:

$$f_n \circ j_n = \mu_E \circ (f_{n-1} \otimes f_1) \circ (\mathrm{id} \otimes i) = \mu_E \circ (f_{n-1} \otimes \eta_E) = f_{n-1}.$$

Finally, the f_n assemble to a map $f \colon MU \to E$, unique up to homotopy. So, all orientations come from a map between homotopy ring spectra $MU \to E$ constructed by the above procedure. In fact, it is also injective, but we won't show it here.

□

Example 1.22.

(1) The orientation of $H\mathbb{Z}$ gives a ring map $MU \to H\mathbb{Z}$ [2, Ex. 9.5.8]. Since MU is connective with $\pi_0 MU = \mathbb{Z}$, this is the truncation $MU \to \tau_{\leq 0} MU$.

(2) The orientation of KU from Talk 1 gives a ring map $MU \to KU$, classically known as the *Todd genus* [2, Ex. 9.5.8, Thm. 11.7.4].

Remark 1.23 (The E_∞ universal property, [4, Rem. 2.33], [5], [6])**.** Since MU is an E_∞-ring ([Remark 9.13] [1], [3, §10.2]), one can ask for a universal property among E_∞-rings. It exists, but it is a different statement: for an E_∞-ring R,

$$\mathrm{Map}_{\mathrm{CAlg}}(MU, R) \simeq \{E_\infty\text{-trivialisations of } BU \xrightarrow{J} \mathrm{Pic}(\mathbb{S}) \xrightarrow{R\otimes -} \mathrm{Pic}(R)\}.$$

This asks for a trivialisation compatible with $\oplus$ up to all higher coherences, which is much more data than a single class $t \in R^2(\mathbb{CP}^\infty)$.

Theorem 1.21 also tells us what *all* homotopy ring maps $MU \to E$ look like once one orientation t_E is fixed.

Since with a fixed orientation we have an isomorphism $E^*(BU(1)) \cong E^*[[t_E]]$, any other orientation is $t'_E = \sum_{i\geq 0} a_i t_E^{i+1}$ with $a_0 = 1$ and $a_i \in E^{-2i}$ arbitrary. We obtain the degrees as the orientation is in degree 2, and $a_0 = 1$, $a_{-1} = 0$ from the definition of an orientation. So, all maps $MU \to E$ are obtained by the construction of the proof of Theorem 1.21 applied to all orientations t'_E of E.

Considering $E \otimes MU$, which carries two orientations (from E and from MU):

$$MU \otimes \mathbb{S} \text{ and } \mathbb{S} \otimes MU \to MU \otimes MU \xrightarrow{t_E \otimes t_{MU}} E \otimes MU,$$

gives the following.

Proposition 1.24 ([Lemma 11.1] [1])**.** *Let E be complex oriented. Then $E_*MU \cong E_*[b_1, b_2, \dots]$, $|b_i| = 2i$, where the b_i are determined by*

$$t_{MU} = \sum_{i\geq 0} b_i\, t_E^{i+1} \quad (b_0 = 1) \qquad \text{in } (E \otimes MU)^*(\mathbb{CP}^\infty).$$

We will also not prove this, but let us give an equivalent description of the proposition. Two different orientations of $E \otimes MU$ give two different formal group laws over E_*MU, so if we have another (ordinary) E_*-algebra R and a map $E_*MU \to R$, we can pull back the two formal group laws to R. Denote the formal group law coming from a map $E_* \to E_*MU \to R$ by F_E and the one coming from MU by F_{MU}. These two formal group laws are different, but they are isomorphic. Indeed, the power series $t_{MU} = \sum_{i\geq 0} b_i\, t_E^{i+1}$ gives a strict isomorphism of formal group laws over E_*MU, which is pulled back to R along $E_*MU \to R$.

In fact, the converse is also true: given a formal group law F over R and a strict isomorphism $F_E \to F$, we can construct a map $E_*MU \to R$ by sending $b_i \mapsto a_i$ where a_i are the coefficients of the strict isomorphism. This is also justified as each formal group law over R comes from a map $MU_* \to R$ by Quillen's theorem.

1.1. **Quillen's theorem.** Since MU is complex oriented, it has a formal group law (coming from the identity map $MU \to MU$),

$$F_{MU}(x, y) = \sum_{i,j} a_{ij} x^i y^j, \qquad a_{ij} \in \pi_{2(i+j-1)} MU.$$

With a standard grading of the Lazard ring so that the universal coefficient a_{ij} has degree $2(i+j-1)$, we get a graded ring map $\theta\colon L \to MU_*$ classifying F_{MU}.

Theorem 1.25 (Quillen, [Theorem 9.15], [Theorem 11.8] [1])**.** *$\theta\colon L \to MU_*$ is an isomorphism. That is, the formal group law of MU is the universal formal group law.*

Combined with Theorem 1.21: ring maps $MU \to E$ correspond to orientations of E, and on homotopy groups such a map is the map $L \to E_*$ classifying F_E.

1.2. **Outlook: MU_*MU and the moduli of formal groups.** Apply Proposition 1.24 to $E = MU$: $MU_*MU \cong MU_*[b_1, b_2, \dots]$.

By Quillen's theorem, a ring map $MU_* \to R$ is a formal group law over R; by Proposition 1.24, extending it to MU_*MU is the choice of a strict isomorphism to another formal group law. Hence:

Theorem 1.26 ([Theorem 11.10] [1])**.** *The pair $(MU_*, MU_*MU) \cong (L, L[b_1, b_2, \dots])$ is a cogroupoid object in commutative rings, corepresenting the functor*

$$R \longmapsto \{\textit{formal group laws over } R \textit{ and strict isomorphisms between them}\}.$$

Remark 1.27. Corepresentability is meant in the following sense:

The functors $\mathrm{Hom}_{\mathrm{coGrpd(CRing)}}((MU_*, MU_*MU), -)$ and $\mathcal{FGL}^s$ are naturally isomorphic, where $\mathcal{FGL}^s$ denotes the groupoid of strict isomorphisms between formal group laws, and rings are considered as discrete cogroupoids.

A cogroupoid object means that there is a diagram of rings:

$$MU_* \underset{\longleftarrow \varepsilon \longrightarrow}{\overset{\mathrm{id}\otimes 1}{\longrightarrow}\ \ \ \